7.4 Adding and Subtracting Rational Expressions

Name _____ Period ____ Date _____

Algebra II 7.4 (30-minute plan) · Student sheet

I can…

- add and subtract rational expressions with like denominators.
- find the least common denominator and add or subtract with unlike denominators.
- simplify complex fractions.

Vocabulary · fill in the missing word

Least common denominator (LCD) — the least common _____ of the denominators

Complex fraction — a fraction with a fraction in its numerator, its _____, or both

Part 1 · Notes and examples (fill in each note, then work the example)

Like denominators: Like denominators: add the _____; keep the _____.

Subtracting: When subtracting, put the second numerator in _____ and distribute the minus sign.

Example 1

Find the difference: $\frac{5}{x+2} - \frac{x+3}{x+2}$

Least common denominator: LCD: _____ each denominator and use each factor to its _____ power.

Example 2

Find the least common multiple of $4x^2 - 16$ and $6x^2 - 24x + 24$.

Complex fractions: Complex fraction: multiply the top and bottom by the _____ of every small fraction.

Example 3

Simplify: $\left(\frac{5}{x+4}\right) \div \left(\frac{1}{x+4} + \frac{2}{x}\right)$

Part 2 · Partner practice

Partner practice

Find the sum: $\frac{2}{3x} + \frac{1}{6x^2}$